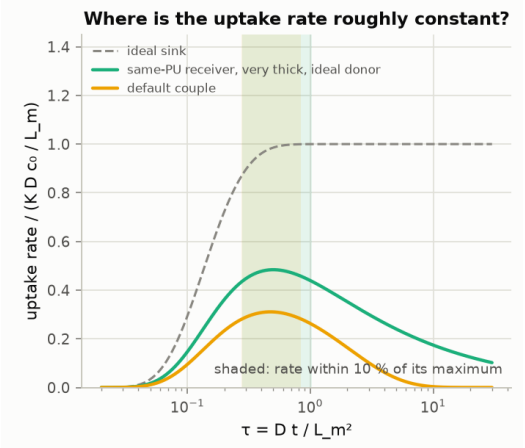
Computational design toolkit for a polymer electrolyte experiment (Li–S batteries)

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Motivation. The planned experiment compares seven polyurethane (PU) solid electrolytes with one HDI/1,4-butanediol hard segment and soft segments that change one variable at a time (four polyethers with C2, C3 or C4 spacers or a methyl branch, two polycarbonates, one polyester), to separate the solubility, mobility and chemical consumption of lithium polysulfides in the polymer. Before any polymer has been made, I used computation to design the key transport experiment, to build and test the analysis of the data to come, and to write down testable predictions with a fixed decision rule.

Diffusion-couple model: designing the transport experiment

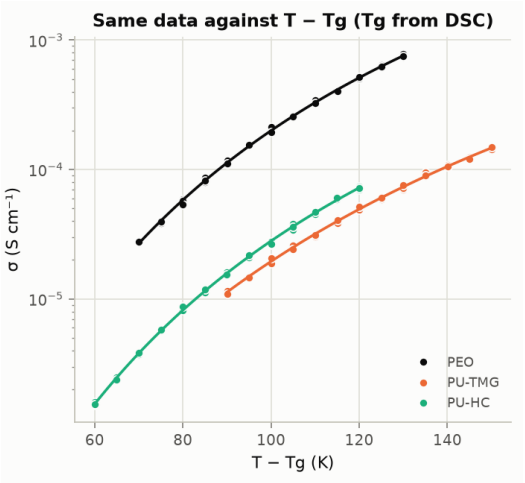
Problem: which sampling times and geometry make the polysulfide partition coefficient K and diffusivity D measurable in the planned PEO-donor | PU-membrane | PU-receiver couple? *Method:* a 1-D finite-volume model verified against an exact Laplace-domain solution, with Fisher-information design, Bayesian inversion (emcee) and global sensitivity analysis. *What it changed:* the PU receiver is not an ideal sink (peak uptake rate 0.31–0.48 of $K \cdot D \cdot c_0/L$), so the slope method would underestimate the permeability and a full-model fit is used instead; the planned first sample at 0.1 L^2/D is below quantification in 22 of 28 scenarios; K and D separate only with a near-equilibrium sample or Raman profiles (otherwise their posterior correlation is −0.95 to −0.98); and the UV-vis detection limit is measured first.



Receiver uptake rate against dimensionless time (assumed inputs); dashed: ideal sink.

Analysis pipeline with uncertainties

Problem: the impedance, polarisation, DSC, UV-vis, FTIR, diffusion-couple and cell data must become comparable numbers with uncertainties. *Method:* an automated CSV-based pipeline (equivalent-circuit fits, VFT, Bruce–Vincent F^+, saturation limit, couple inversion, cell metrics) with Monte Carlo error propagation, tested on synthetic data with known truths (27 of 37 quantities recovered within 3 %). *What it changed:* the tests made the impedance fits modulus-weighted and the saturation limit subject to a significance test, and showed that the low-R Li_2S_8 films need LiTFSI reference films at the same R.

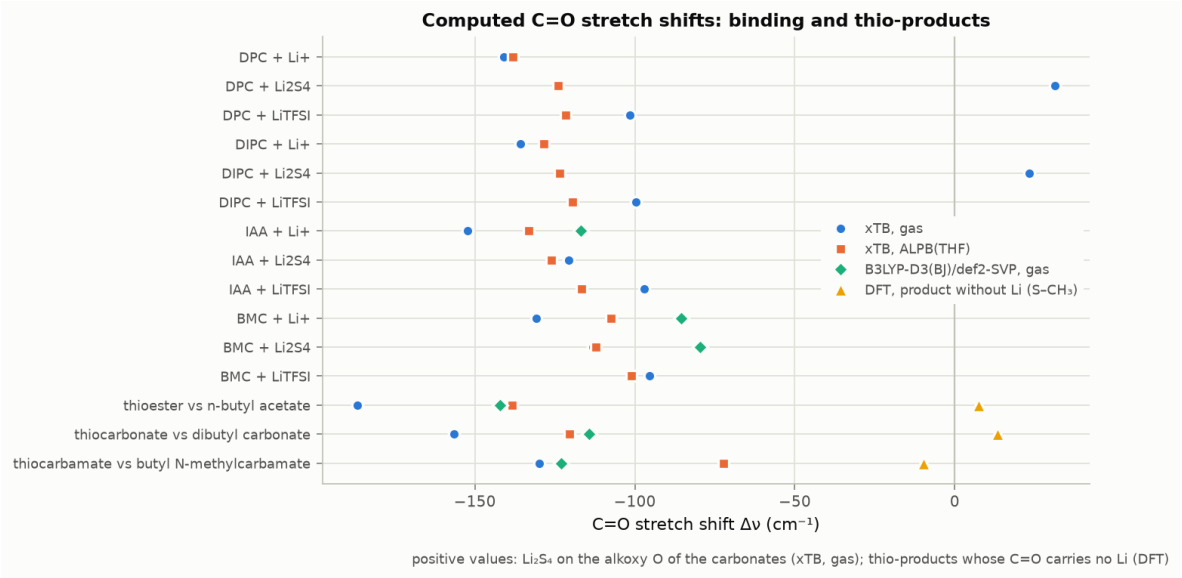


Synthetic test data: σ against $T - T_g$ (T_g from automated DSC); lines: VFT fits.

Molecular calculations: xTB screening with DFT checks

Problem: do model molecules of the soft segments bind Li^+ and polysulfides differently enough to matter, and where would reaction products appear in FTIR? *Method:* GFN2-xTB with many starting structures (gas phase and implicit THF), checked by counterpoise-corrected B3LYP-D3(BJ) single points, and DFT harmonic frequencies. *What it changed:* xTB does not resolve binding differences of a few kJ mol^{-1} (DFT confirmed 1 of 4 above-threshold xTB differences where differences were small, 8 of 8 where they were large), so such differences are not used; a thio-product C=O band lies 114–142 cm^{-1} below the parent band only when Li^+ sits on it and otherwise within 14 cm^{-1} of it, under the polymer's own C=O band, so the FTIR test gets an un-aged control and products are confirmed by NMR of model compounds.

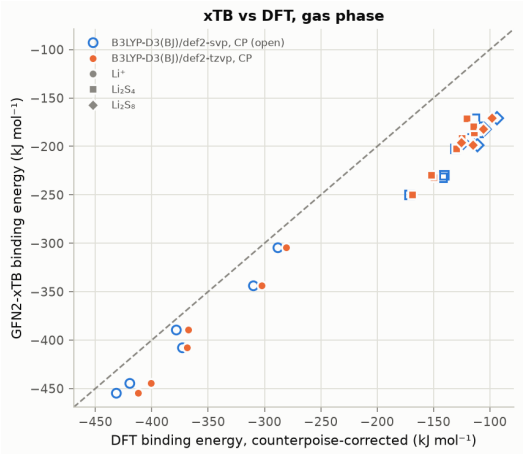
Full code available on request.



Computed C=O stretch shifts on binding Li^+, Li_2S_4 or LiTFSI and for the thio products (xTB: gas phase and implicit THF; DFT: gas phase).

Predictions written down before the experiment

Problem: can these calculations predict the experiment at all? *Method:* before the first synthesis I fixed 43 rule-based entries (22 with a predicted direction) that link computed to measured quantities, each with a threshold or interval, a decision rule (supported / not supported / inconclusive against the combined 1σ uncertainty) and a scoring script. *What it changed:* in version 2, still before any measurement, three-oxygen ether models became the primary models, and every polysulfide-based direction must also hold in counterpoise-corrected DFT with the same sign and above the threshold; 7 xTB directions lacked this confirmation and became “no directional prediction”.



GFN2-xTB against counterpoise-corrected DFT binding energies (gas phase); dashed: equality.

Methods & tools: finite-volume model with Laplace-domain verification; emcee Bayesian inversion and identifiability; global sensitivity analysis; GFN2-xTB; B3LYP-D3(BJ) with counterpoise correction; harmonic frequencies; pytest (61 tests), seeded runs.

Limitations: all inputs are assumptions and there are no experimental data yet; single model molecules cannot represent the polymer (chain cooperativity, hydrogen bonding, microphase separation); DFT was used at validation level only (gas-phase single points on xTB structures; frequencies with def2-SVP). The code was written with the assistance of the AI coding assistant Claude Code (Claude Opus 5.5, max effort); the scientific questions, mappings and decision rules were set by me.

Full code available on request.